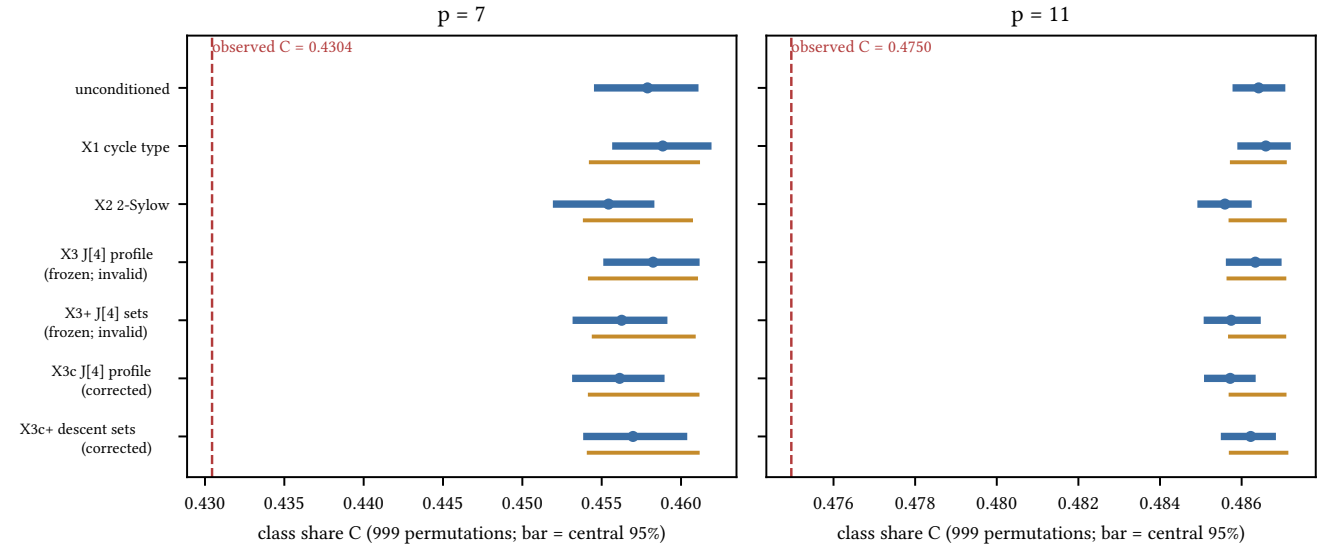
Run report: class invariants, p = 7 and p = 11

Study `research/smith_class_invariant_20261015`. Code at `0bff9255b` (both primes). Host: EC2, Amazon Linux 2023, 2 vCPU, 7 GB, no isolation (L0); timings are descriptive only. Commands: `smith_class_invariant compute --p 7|11 --out data`, then `smith_class_invariant summarize --dir data 7 11`. Started 2026-10-15T23:32Z. Inputs: frozen `research/smith_iso_granularity_20261015/data/iso_p{7,11}.csv.gz`.

recorded	**p = 7**	**p = 11**
git head	0bff9255b	0bff9255b
isomorphism classes / tractable (computed)	34227 / 16823	324562 / 160010
X2 uncertified classes	0	0
K2 / K3 / K4 / K4′ / K5 / K5c	`true` `true` / `true` / `false` / `true` / `false` / `true`	`true` `true` / `true` / `false` / `true` / `false` / `true`
K4 mismatching classes (frozen descent)	4405	43538
seconds, threads	238, 2	11287, 2
observed C; unconditioned null 2.5–97.5%	0.4304 0.4545–0.4611	0.475 0.4858–0.4871



blue: null conditioned on (stratum, invariant), dot = median; orange: L-class placebo of equal cardinality. Source data/summary_p*.json.

Figure 1: Recorded values of this run; see `../../REPORT.pdf`.

Both compute runs completed; none failed or timed out. K4 and K5 fail for the frozen descent (X3, X3+), as recorded in Amendment 1; these candidates are reported invalid. Raw records: `data/inv_p{7,11}.csv.gz`, `data/controls_p*.json`, `data/summary_p*.json`, `data/run_p*.log`.